Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Acceleration

Meaning in my words How would you explain acceleration to a friend?	**How can you tell?** A cart follows a bend at steady speed. Why is it accelerating?
Example or sketch Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits?	**Non-example and why** A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating?

Speeding up

Meaning in my words How would you explain speeding up to a friend?	**How can you tell?** A cart changes from −2 to −5 m/s. What happens to its speed?
Example or sketch Draw three equal-time dots for a cart speeding up to the right.	**Non-example and why** A cart moves right with acceleration to the left. Is it speeding up before it turns?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slowing down

Meaning in my words How would you explain slowing down to a friend?	**How can you tell?** Velocity changes from −5 to −2 m/s. What happens to the speed?
Example or sketch Draw velocity and acceleration arrows for a bike braking while moving right.	**Non-example and why** “Negative acceleration always means slowing down.” What happens if velocity is negative too?

Turning

Meaning in my words How would you explain turning to a friend?	**How can you tell?** A rider keeps the speedometer at 4 m/s around a bend. Which part of velocity changes?
Example or sketch How would you draw two velocity arrows as a cart turns from east to north at steady speed?	**Non-example and why** A cart brakes on a straight line and still moves east. Has it turned?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Before

After

My marks mean

Choose the speed alert

07

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity change to find acceleration.

CHOOSE ONE JOB TOGETHER

TELEMETRY ENGINEER ☐

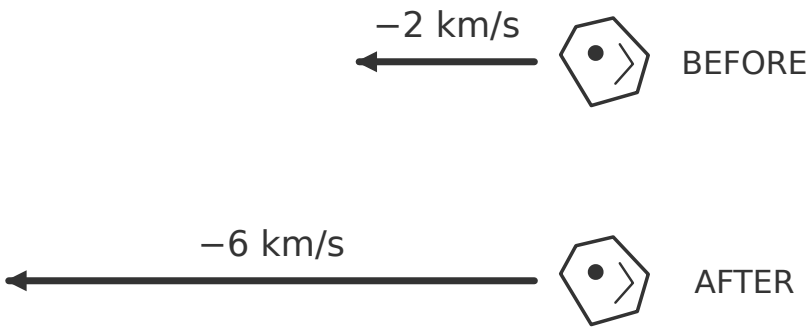
Radio the average acceleration with its sign and unit. Label both velocity arrows so the crew can check it.

REPLAY COMMENTATOR ☐

Narrate the two-second change. Point to the arrow lengths to explain whether XJ sped up or slowed down.

Point to a clue. Keep your first marks; add the update using the color key.

Two signed XJ velocities are **−2** and **−6 km/s**, two seconds apart. The control room must choose its speed alert.



01 OUR FIRST CALL

Which speed alert should the control room use? Support it with the arrows.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the proposed banner. Give a corrected report using the two velocities.

Switch speakers. Which arrow shows the larger speed?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks